

Test Plan and Cases

Test Plan & Test Cases — OptiCrop

1. Testing Objectives

- Verify model prediction accuracy across the compared algorithms
- Verify Flask application correctly handles valid and invalid inputs
- Verify end-to-end flow from form submission to result display

2. Model Evaluation

Model	Metric(s)	Notes
KNN	Accuracy, Precision, Recall, F1-score	Sensitive to choice of K
Logistic Regression	Accuracy, Precision, Recall, F1-score	Baseline linear model
Decision Tree	Accuracy, Precision, Recall, F1-score	Prone to overfitting; monitor depth
Random Forest	Accuracy, Precision, Recall, F1-score	Typically most robust
K-Means	Silhouette Score	Used for exploratory clustering, not final prediction

Best-performing model (typically Random Forest or KNN on standard crop datasets) selected and saved as `model.pkl`.

3. Functional Test Cases

Test ID	Description	Input	Expected Result
TC-01	Valid input submission	N=90, P=42, K=43, temp=20.8, humidity=82, pH=6.5, rainfall=202	Returns a valid crop recommendation
TC-02	Missing field	One field left blank	Form shows validation error, no prediction made
TC-03	Non-numeric input	Text entered in numeric field	System rejects input with error message
TC-04	Out-of-range value	Negative rainfall or pH > 14	System flags input as invalid
TC-05	Boundary values	Min/max plausible values for each parameter	System processes without crashing
TC-06	Model load failure	<code>model.pkl</code> missing/corrupted	System shows graceful error, not a crash
TC-07	UI responsiveness	Access form on mobile browser	Form displays correctly and is usable
TC-08	Prediction speed	Standard valid input	Result returned within 2 seconds

4. Test Environment

- Flask development server
- Browser-based manual testing (Chrome/Firefox)
- Local machine meeting minimum hardware requirements (i3, 4GB RAM)

5. Test Summary

All functional test cases should pass before deployment. Model accuracy metrics should be documented and compared across all five algorithms, with justification for the final model selected.